Development and internal validation of a machine learning model using Python to predict severe knee injuries in NBA players

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Background & objective

Among athletes in major sports, basketball players face a higher risk of injury due to repetitive musculoskeletal stresses, especially from pivoting, jumping, and changing direction. Knee injuries are among the most common injuries in professional basketball players (Tummala et al., 2022).

The purpose of this project is to predict the risk of a serious season-ending knee injury based on a player's in-game statistics during an NBA season.

Methods & Results

I followed the seven steps described by Steyerberg & Vergouwe (2014) and Shipe et al. (2019) for developing valid prediction models, as shown in Table 1 and briefly explained.

Table 1. Steps for developing valid prediction models

Step	Relevant questions
(1) Problem definition and data inspection	What is the precise research question? Are the aim predictors or predictions? What is the endpoint of interest? What are the candidate predictors? What are the data sources?
(2) Coding of predictors	Categorical or continuous predictors? Missing values? Multicollinearity?
(3) Model specification	Which predictors to select for the model? Model form/equation? Assessment of assumptions? (e.g. linearity)
(4) Model estimation	Is there sufficient data for model estimation? Which method to use to estimate the model coefficients or parameters from the data?
5) Model performance	What aspects determine predictive performance? What are appropriate measures? Does the model work well on the available data?
(6) Model validity	How to validate the model's performance? Is it expected to work well in other similar datasets?
(7) Model presentation	How to present the model in a formula or a score chart?

Step 1: Problem definition and data inspection

In this project, I was interested in predicting the probability of a severe knee injury during an NBA season that results in a player missing the rest of the season. I was interested in developing a multivariable prediction model from some combination of the following widely available variables:

- Age,
- Position (Point guard, Shooting guard, Small Forward, Power Forward, Center),
- Number of games played and/or started,
- Minutes played (average per game),
- Number of two-point, three-point and field total attempts made (averages per game),
- Number of offensive, defensive and total rebounds made (averages per game),
- Number of blocks and personal fouls (averages per game),
- Number of assists, steals made (averages per game).

My research question was whether the information above, when combined, could accurately predict the probability of a severe, season-ending knee injury.

Data sources

I used a combination of two datasets that I found on Kaggle:

- Injury dataset: The first dataset included details on every injury in the NBA from the beginning of the 2010-2011 season through the end of the 2019-2020 season. The data was scraped from the Pro Sports Transactions website (https://www.prosportstransactions.com).
- In-game statistics dataset: The second dataset included player in-game and performance statistics extracted from https://www.basketball-reference.com.

Data processing, cleaning and linkage

Injury dataset:

The dataset contained records of various types of injuries. I was interested in knee-related injuries, so I wrote a function that scans each row to identify keywords such as “knee”, “meniscus”.This is what the initial dataset looked like.

No.	dateddmmyyyy	team	playername	notes
20	4/4/2015	Hornets	Al Jefferson	sore right knee (out for season)
21	12/30/2015	Hornets	Al Jefferson	torn lateral meniscus in right knee (out indefinitely)
22	12/31/2015	Hornets	Al Jefferson	surgery on right knee to repair torn lateral meniscus (out indefinitely)

23	1/1/2016	Hornets	Al Jefferson	placed on IL recovering surgery on right knee to repair torn lateral meniscus
24	9/25/2017	Suns	Alan Williams	surgery on right knee to repair torn meniscus (out indefinitely)
25	10/18/2017	Suns	Alan Williams	placed on IL recovering from surgery on right knee to repair torn meniscus

For season-ending injuries or injuries that kept the athlete out indefinitely, the injury list typically noted this. So, I classified players as 1 in a new “Severe Knee Injury” column when the notes contained the phrase “out indefinitely” or “out for season,” and 0 for less serious injuries. I also used the date of injury to classify each record for each year of the NBA season (2010-2011, 2011-2012, …, 2019-2020), which starts in October. Some players had many rows for the same season, listing the same injury. I believe that this was for injuries where the team added them to the inactive list (IL) and listed every game they missed. I dropped these duplicates based on player name, team, and season. Finally, some cleaning in team names was required as a few teams changed names over time (e.g. New Orleans Pelicans and Hornets), and I needed to standardise the team names between this dataset and the game-statistics data set (one dataset was referring to a team using the city name, the other did not, e.g. Boston Celtics vs. Celtics).

This is what the final injuries dataset looked like:

No.	season	teamFN	playername	svrkneeinjur	severity
1	2010-2011	Hornets, Pelicans, or Bobcats	Gerald Henderson Jr.	0	Placed on IL
2	2010-2011	Milwaukee Bucks	Michael Redd	0	Placed on IL
3	2010-2011	Portland Trail Blazers	Elliot Williams	1	Out for season
4	2010-2011	Phoenix Suns	Robin Lopez	0	Placed on IL
5	2010-2011	Oklahoma City Thunder	Kevin Durant	0	Placed on IL
6	2010-2011	Washington Wizards	Andray Blatche	0	Placed on IL
7	2010-2011	Boston Celtics	Shaquille O'Neal	0	Placed on IL
8	2010-2011	Oklahoma City Thunder	Nick Collison	1	Out indefinitely
9	2010-2011	Orlando Magic	Vince Carter	0	Placed on IL
10	2010-2011	Boston Celtics	Kendrick Perkins	0	Placed on IL

In-game statistics dataset:

Here’s what the initial dataset looked like after removing game statistics I considered irrelevant to the prediction problem (e.g., total points scored, number of turnovers, percentages of successful field goal attempts, etc.).

season	playername	teamFN	pos	age	g	gs	mp	fga	p3a	p2a	fta	orb	drb	trb	ast	stl	blk	pf
2015-2016	A.J. Hammons	Dallas Mavericks	C	24	22	0	7.4	1.9	0.5	1.5	0.9	0.4	1.3	1.6	0.2	0	0.6	1
2014-2015	A.J. Price	Cleveland Cavaliers	PG	28	11	0	7.9	3.1	1	2.1	0.5	0.4	1	1.4	1.2	0.3	0	0.1
2014-2015	A.J. Price	Indiana Pacers	PG	28	10	0	19	8.9	3.9	5	1.8	0.2	1.2	1.4	2.7	0.4	0	0.9
2011-2012	A.J. Price	Indiana Pacers	PG	25	44	1	13	4	2	2	0.8	0.3	1.1	1.4	2	0.5	0	0.7
2010-2011	A.J. Price	Indiana Pacers	PG	24	50	0	16	6.4	3	3.4	1.6	0.3	1.1	1.4	2.2	0.6	0	1.2

Abbreviations: pos "Position", age "Age (years)", g "Number of games played", gs "Number of games started", mp "Minutes played (avg.)", fga "Field goals attempts (avg.)", p3a "3-point field goal attempts (avg.)", p2a "2-point field goal attempts (avg.)", fta "Free throw attempts (avg.)", orb "Offensive rebounds (avg.)", drb "Defensive rebounds (avg.)", trb "Total rebounds (avg.)", ast "Assists (avg.)", stl "Steals (avg.)", blk "Blocks (avg.)", pf "Personal fouls (avg.)".

This dataset also required data cleaning to standardise team names, remove odd characters from player names (such as á, Á, ã, ā, ç, č, etc.) that were not present in the injury dataset, and deduplicate multiple records for players who had been traded during a season.

Linkage:

I joined the injuries dataset to the player stats dataset by matching player name, season, and team. The final dataset contained n = 5010 player records, of which 234 (4.7%) were season-ending knee injuries.

Step 2: Coding of predictors

Categorical or continuous predictors

Most predictors utilized by the model are linear, except the variable representing a player's position, which is a categorical variable. To address this, I created 4 'dummy' binary variables that took on values 1 or 0 if the player played in a certain position or not. Notice that 4 variables, instead of 5, were used to avoid position variables completely predicting each other.

Missing values

There were no missing values for any of the predictor variables in the dataset.

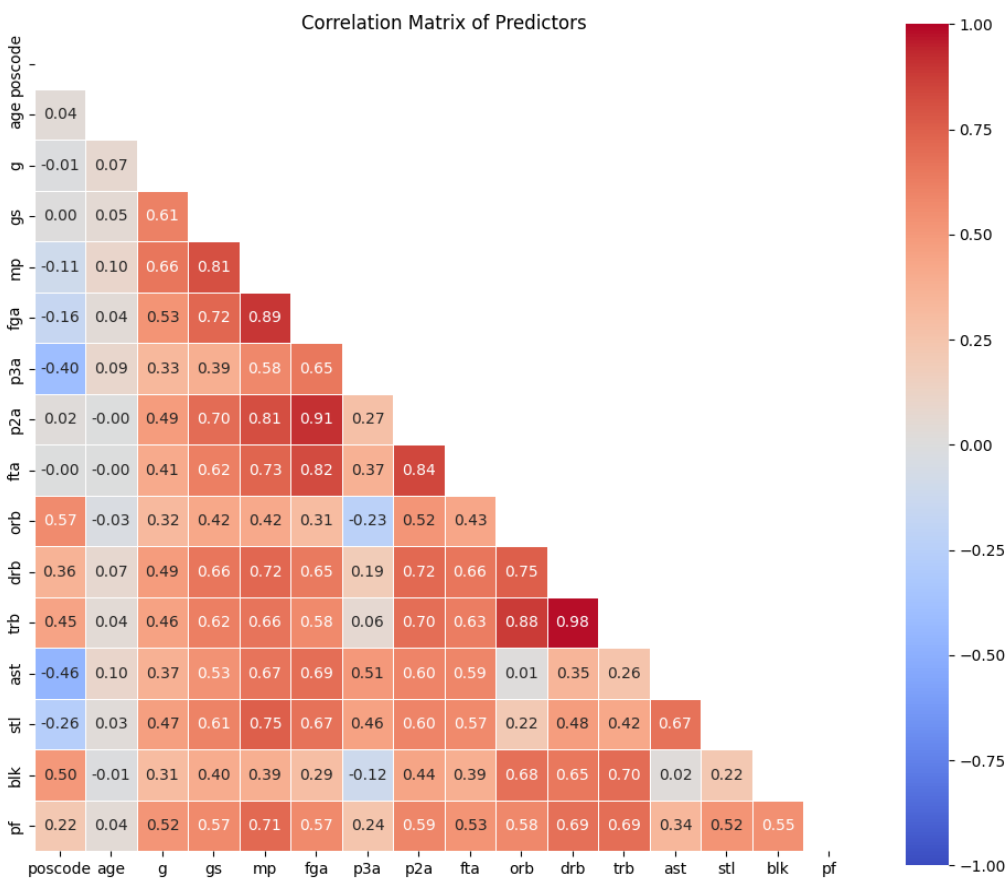
Multicollinearity

The problem is that as predictors become more highly correlated, it becomes harder to determine which predictor actually causes the effect on knee injury risk. Predictors that are highly correlated with each other cannot improve predictions as much as they would if they were not correlated, but still separately correlated with the outcome variable. If two predictors are highly correlated, neither is doing much more work than the other is already doing and would do on its own. This can happen if the predictors capture the same underlying construct, in which case neither adds much more than the other.

Multicollinearity can inflate the variance of the coefficients in a predictive model, potentially preventing it from capturing the most influential predictors (Schreiber-Gregory 2017, Daoud 2017). The model might overfit and be less likely to generalize to out-of-sample data. Multicollinearity can be resolved by omitting a variable from the analysis that is associated with other variable(s) highly (Schreiber-Gregory 2017, Daoud 2017).

To visualize and easily identify highly correlated variables, I created the correlation matrix shown in Figure 1 below. It is clear that certain variables are highly correlated, which can be problematic. As expected, the drb (defensive rebounds) and orb (offensive rebounds) are highly correlated with the trb (total rebounds) variable; therefore, I dropped the drb and orb variables from the list of candidate predictors. Also, fga (field goals attempted) is highly correlated with p2a (two-point attempts) and p3a (three-point attempts), as the reason is once again obvious. It is worth noting that the variable mp (minutes played) also shows strong correlations, but I kept it in because the model fit the data better with it included (as I later tested).

Figure 1. Correlation matrix of candidate predictors



Step 3: Model specification

Candidate predictor features

After dropping the fga, orb and drb collinear variables, the predictors are as follows:

- Position Center (Position_C)
- Position Power Forward (Position_PF)
- Position Small Forward (Position_SF)
- Position Shooting Guard (Position_SG)
- Age (age)
- Games (g)
- Games started (gs)
- Minutes played per game (mp)
- Number of two-pointers attempted per game (p2a)
- Number of three-pointers attempted per game (p3a)
- Number free throws attempted per game (fta)
- Rebounds per game (trb)
- Assists per game (ast)
- Steals per game (stl)
- Blocks per game (blk)
- Personal fouls per game (pf)

Model form

As I am interested in predicting a binary outcome, I decided to use Logistic Regression as the model's algorithm. Logistic regression is a linear model that uses a sigmoid function to calculate the actual probability. First, it calculates a raw score (LOG_{ODDS}) using a linear combination of the predictor variables (an additive model), where ODDS = P / (1 - P) and P is the probability of severe knee injury. The model takes every variable (Age, Position, etc.) and multiplies it by a coefficient (weight):

LOG_{ODDS} = z = Intercept + (b_{age} • age) + ($b_{position_c}$ • position_c) + … + (b_{pf} • pf)

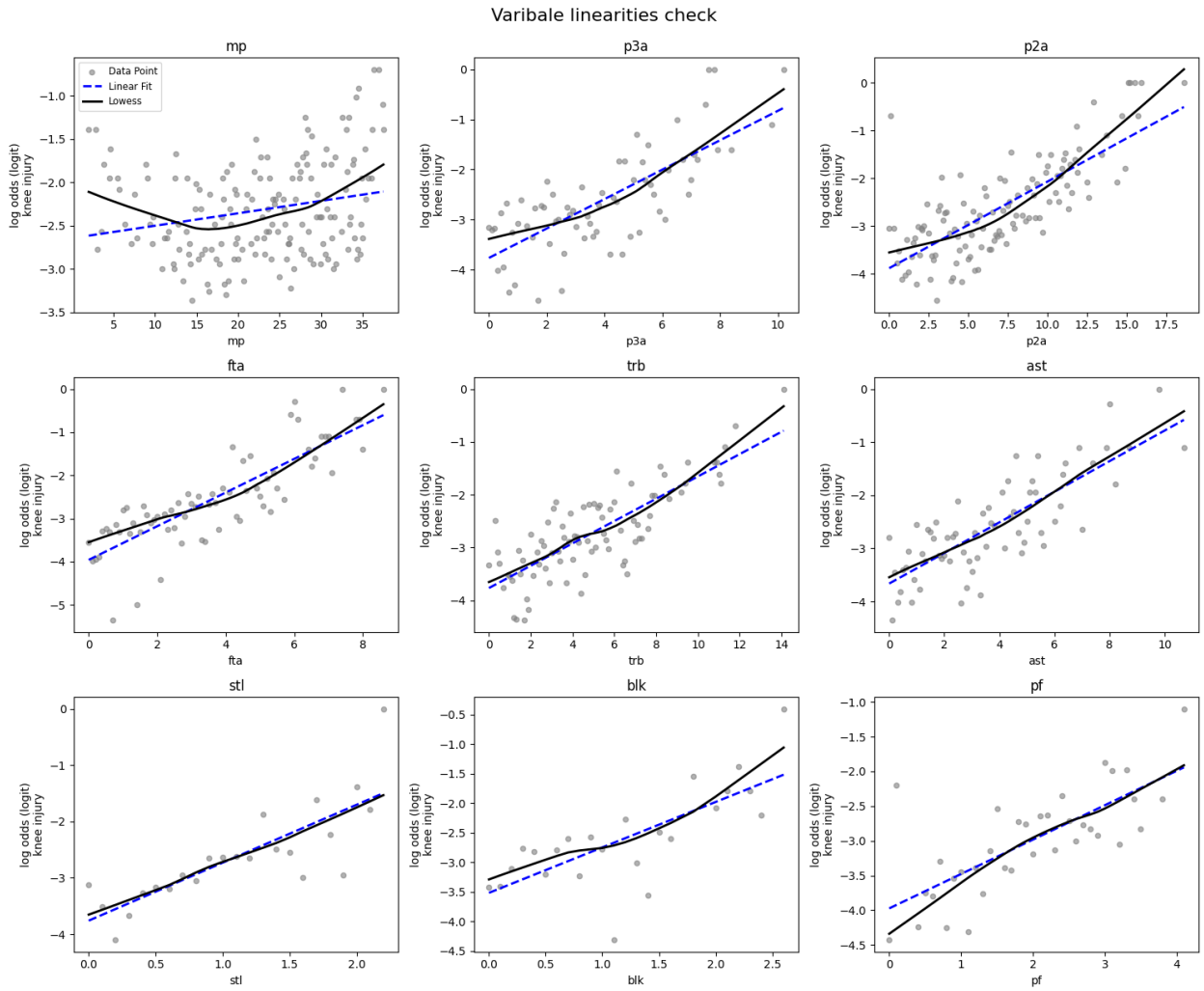
Then it converts the LOG_{ODDS} score into a probability using the sigmoid formula:

$$probability\ of\ severe\ knee\ injury = \frac{1}{1 + e^{z}} = \hat{y}_i$$

Assessment of assumptions

The logistic regression model assumes a linear relationship between the log ODDS of knee injury and each continuous predictor variable (e.g. OG_{ODDS} = a + b_{age} • age). To check the plausibility of this assumption for the data at hand, I constructed the diagrams below:

Figure 3. Graphical assessment of the linearity in the log odds of knee injury with candidate predictor values



We can see that most predictors appear to have approximately linear relationships with the odds of knee injury (on a log scale), except for mp (minutes played), which appears to follow a curve. This could be modelled using a polynomial term, such as $b_1 \bullet mp + b_2 \bullet mp^2$).

Nevertheless, the overall trend in the data points seems to be upwards, and I decided to keep mp in its original numeric scale (as b1•mp) to avoid overfitting, i.e., fitting a model that is too complex and less likely to generalize to out-of-sample data.

Step 4: Model estimation

Method to estimate the model coefficients from the data

The model initially guesses the coefficients, then calculates the cross-entropy loss function J(θ), which is the negative average log likelihood function:

$$J(\Theta) = \frac{-1}{m}\sum_{i=1}^{m} \quad y_i[log(y\hat{}_i) + (1 - y_i)log(1 - y\hat{}_i)]$$

Where:

m = the total number of players

y_i = the Actual Label (The Truth)

$y\hat{}_i$ = The Predicted Probability

The model minimizes the loss function J(θ) using an iterative optimization algorithm called Gradient Descent. Once the global minimum is found, the model has “converged” and the coefficients are finalized.

Once the model is trained and fitted with the data, we get the following coefficients and intercept:

Predictor	Coefficient
Intercept	-3.85806
Position_SG	-0.495238
Position_SF	0.141105
Position_PF	0.009660
Position_C	-0.047355
Age	-0.003111
Games	-0.023059
Games started	-0.013115
Minutes played	0.053863
P3A	0.081855
P2A	0.060058
FTA	0.035992
Total rebounds	-0.071093
Assists	-0.032491
Steals	0.536037

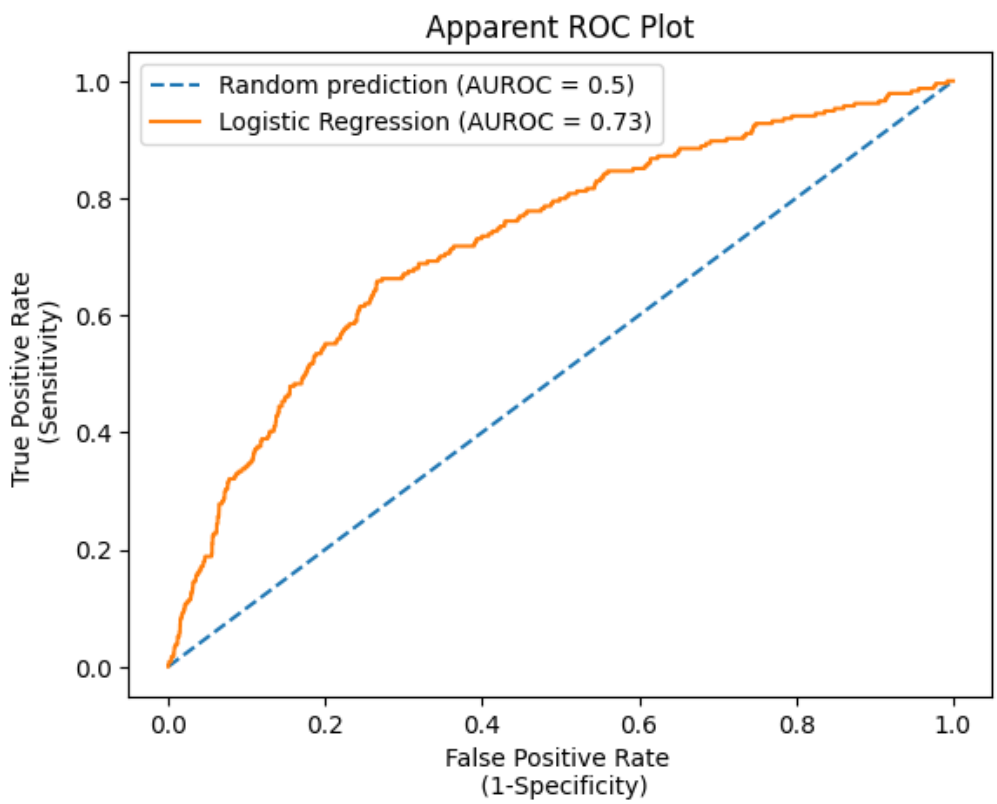
Blocks	0.325016
Personal fouls	0.262671

Step 5: Model performance

Discriminatory ability of the model

To assess the model's performance, I plotted the Receiver Operating Characteristic (ROC) curve and calculated its area under the curve. This is the most common metric for judging how well a binary classification model distinguishes between two classes. The ROC curve is as shown below:

Figure 4. Receiver Operating Characteristic curve for the model estimated using the training dataset



To understand the AUC metric, imagine we select one random player who actually got injured and one who stayed healthy. We ask our model to predict the injury probability for both. The ROC AUC Score is the probability that the model gives a higher risk score to the injured player

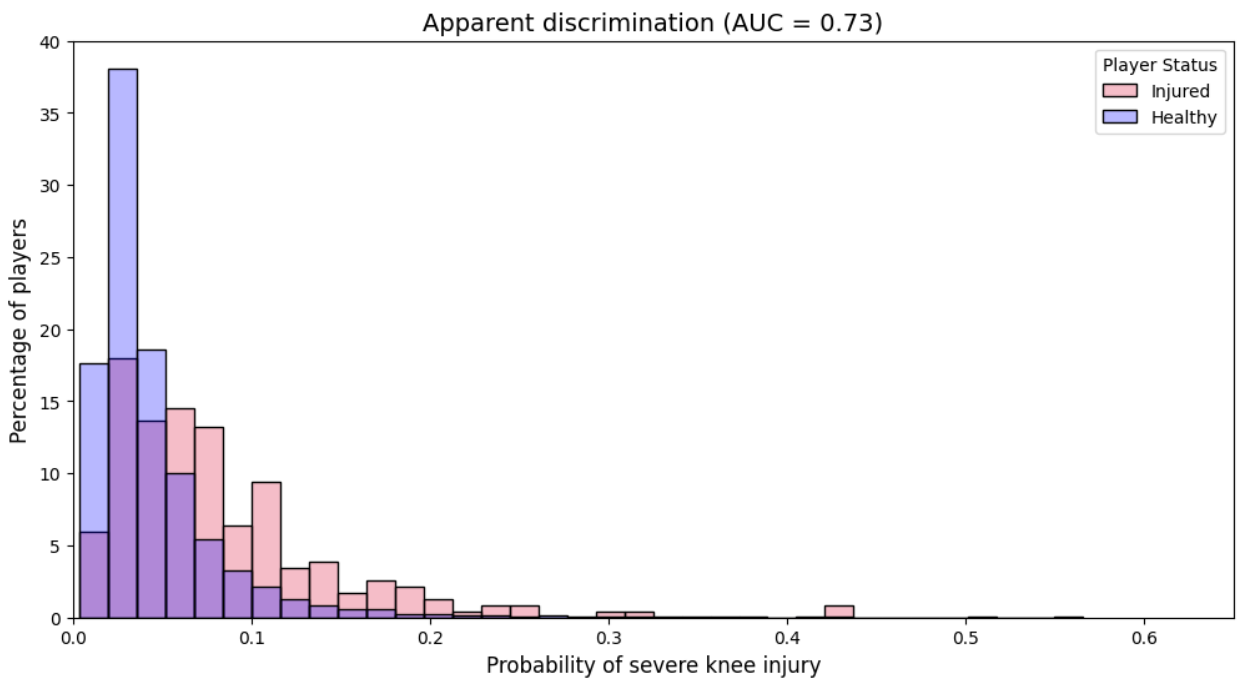
than to the healthy player. Here, the AUC is 0.73, which means: "If I randomly pick an injured player and a healthy player, 73% of the time my model correctly assigns a higher risk to the injured one."

The score always ranges from 0.0 to 1.0, and its usual interpretation is as follows:

- 0.50: Random Guessing. The model is as good as flipping a coin. It has learned nothing.
- 0.50 - 0.70: Poor to Fair. The model has found some signal but makes many mistakes.
- 0.70 - 0.80: Good. This is generally considered a solid, useful model in fields like sports or medicine where noise is high.
- 0.80 - 0.90: Excellent.
- 1.0: Perfect. The model perfectly separates every injured player from every healthy player

I also created a diagram to further visualize the model's discrimination ability:

Figure 5. Discriminatory ability of the model to rank players according to their risk of knee injury



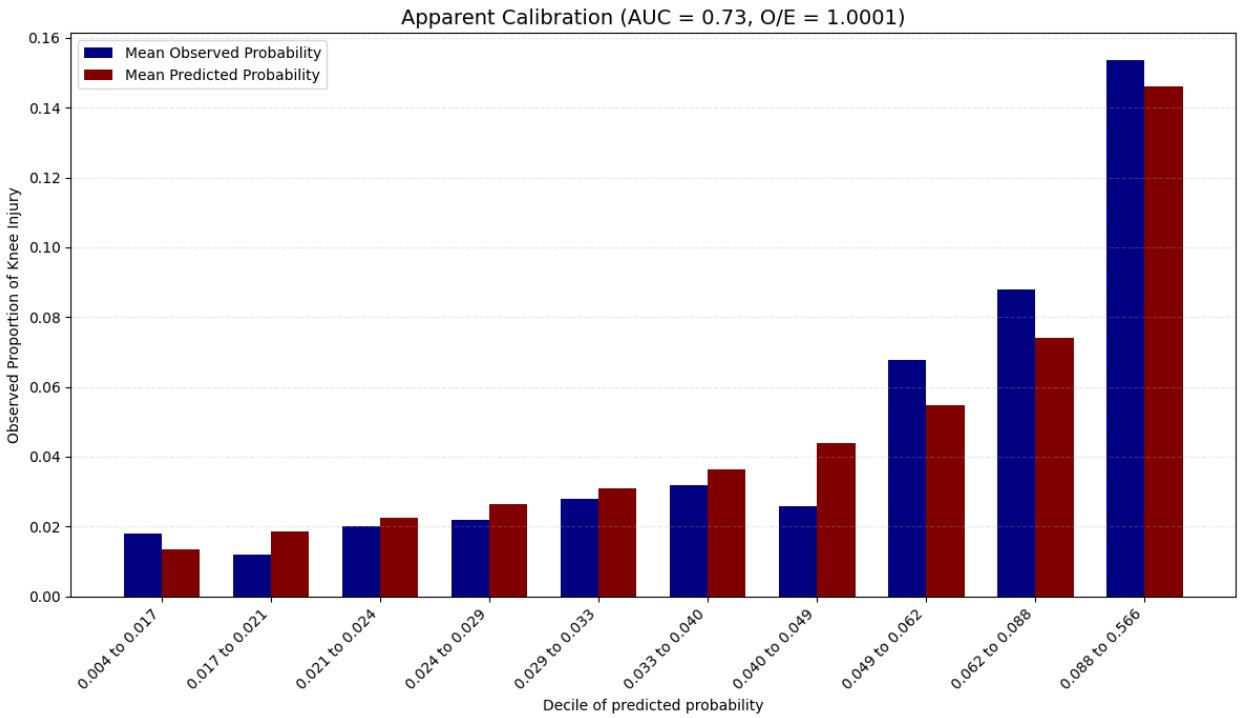
Model Calibration

We should also examine the model's calibration. Model calibration measures how well the model's predicted probabilities align with the observed outcome proportions. If a model predicts a 70% risk of injury for a player with a certain predictor pattern (profile of game-statistics), then if the model is perfectly calibrated, exactly 70 of 100 players with the same predictor profile should actually get injured.

Model calibration can be assessed by examining the calibration bar graph in Figure 6, which compares model-predicted probabilities to observed proportions of knee injury across 10 groups defined by deciles of predicted probability.

Overall calibration was excellent, as the observed-to-expected ratio, i.e. the ratio of the mean observed to the mean predicted probability of knee injury, was O/E=1. From Figure 6, we can see that the agreement between observed and predicted probabilities was quite good across the whole spectrum of predicted probabilities.

Figure 6. Calibration bar plot for the model as estimated in the training data set



Step 6: Model validity

Method to validate the model's performance

The performance of a prediction model on the dataset from which it was derived is likely to be overly optimistic, which is the reason why it is called "apparent" or "naive" performance. Often, model performance appears good on the training dataset but is much lower when evaluated on new data (Steyerberg et al., 2001).

The key is that model performance needs to be examined on data that were not used to build the model initially. This can be done internally, using the same data source, with k-fold cross-validation. K-fold cross-validation evaluates a model by splitting the dataset into 'k' equal parts

(folds); the model is trained on k-1 folds and tested on the remaining one, repeating this process 'k' times so each fold serves as the test set once, with the final performance being the average of all iterations for a more reliable performance estimate.

Is the model expected to work well in other similar datasets?

To answer this, I applied stratified 10-fold cross-validation.The data was split into 10 groups (K=10 folds) using a stratified split that ensured each fold has the same proportion of injuries as the original dataset. Since severe knee injuries are rare (about 3-6% of players per season), using stratified k-fold cross-validation is essential for having an adequate number of injuries in each 10% test.

Figures 7-9 show the cross-validated discrimination and calibration of the model, respectively. The results are slightly worse than before, which is expected, but they do not show a concerning amount of performance loss:

Figure 7. Cross-validated Receiver Operating Characteristic curve

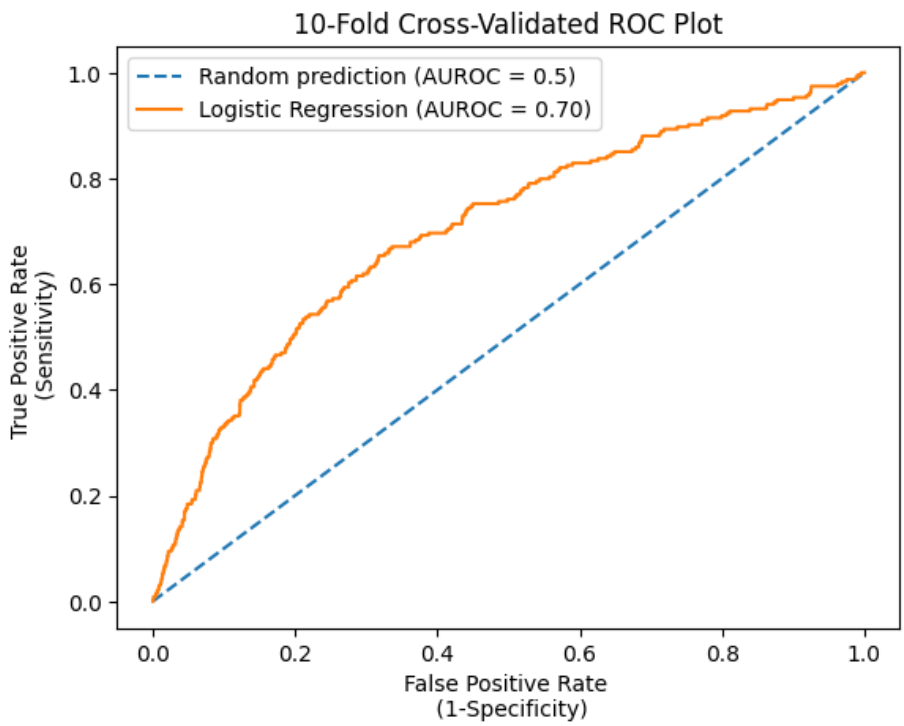


Figure 8. Cross-validated discrimination histogram

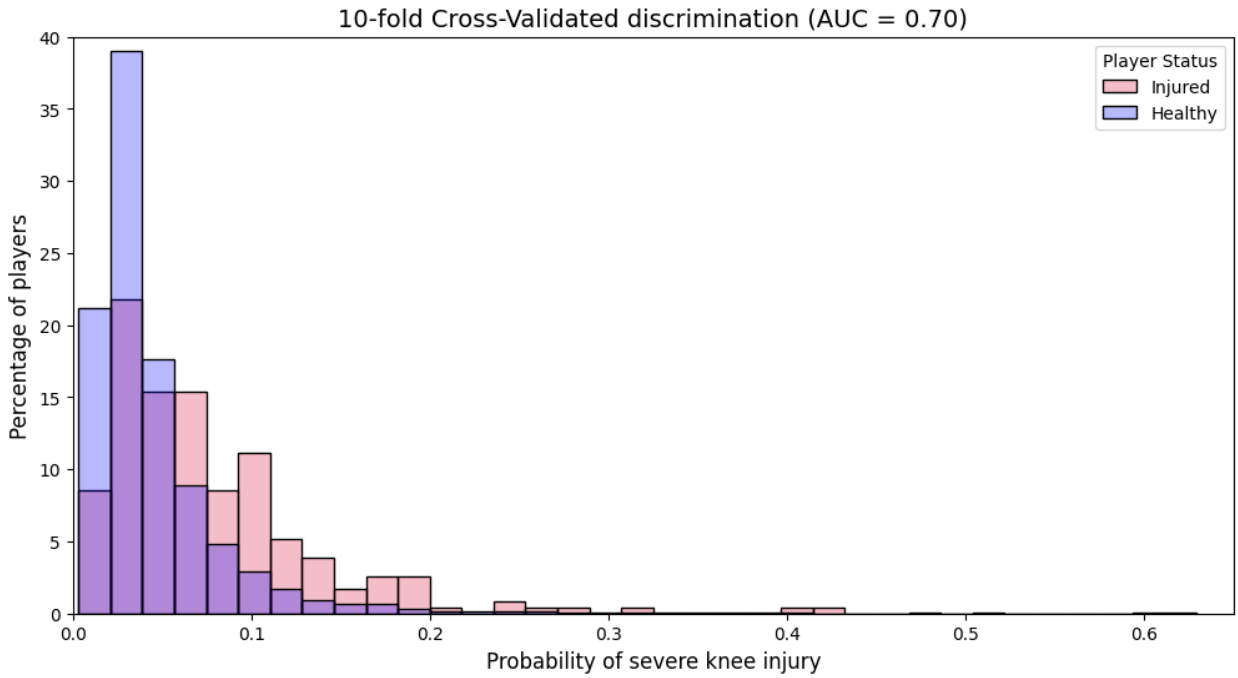
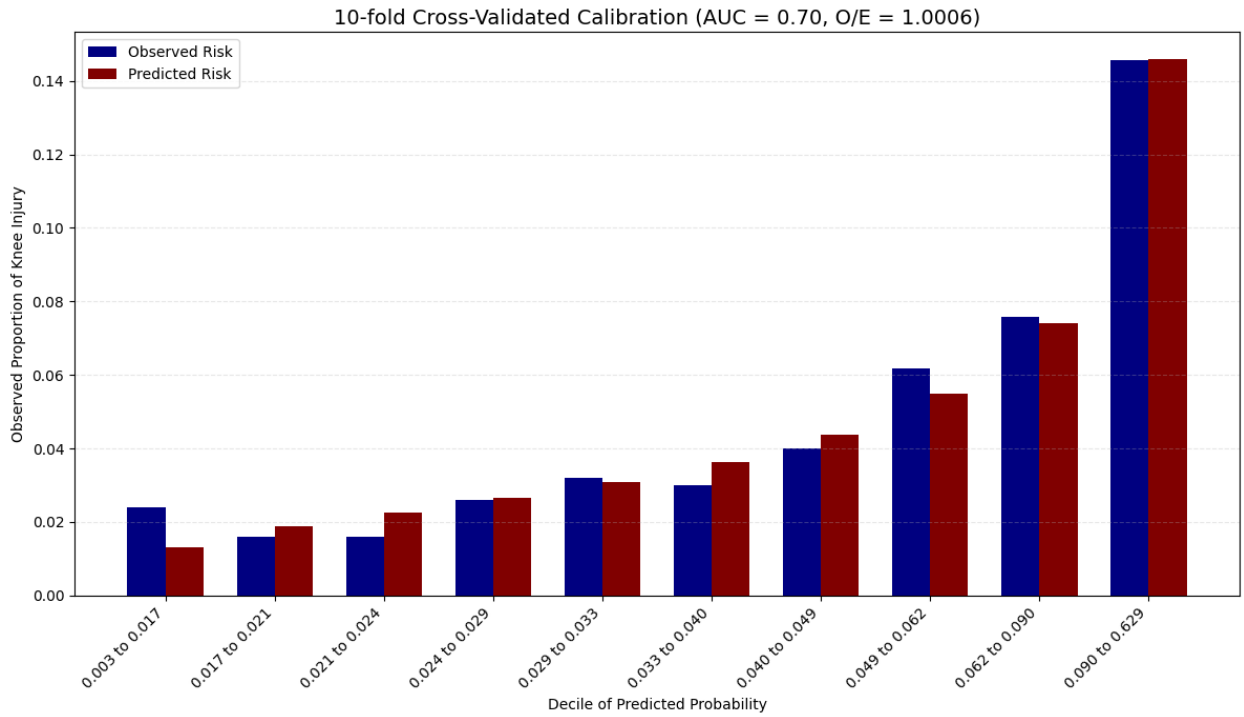


Figure 9. Cross-validated calibration bar plot



Step 7: Model presentation

The model can be presented by writing its equation as shown below. To see how this works in practice, we calculated predictions for two example point guard players in the 2019-2020 season: LeBron James, who did not suffer a knee injury, and Kyrie Irving, who did.

Prediction model		**Example 1: LeBron**		**Example 2:Kyrie Irving**	
Parameter	Coefficient	Parameter values	Coefficient	Parameter values	Coefficient
Intercept	-3.85806	Intercept	-3.85806	Intercept	-3.85806
Position_SG	-0.495238	Point guard	0	Point guard	0
Position_SF	0.141105	Point guard	0	Point guard	0
Position_PF	0.00966	Point guard	0	Point guard	0
Position_C	-0.047355	Point guard	0	Point guard	0
Age	-0.003111	35	-0.108885	27	-0.083997
Games	-0.023059	67	-1.544953	20	-0.46118
Games started	-0.013115	67	-0.878705	20	-0.2623
Minutes played	0.053863	34.6	1.8636598	32.9	1.7720927
P3A	0.081855	6.3	0.5156865	7.1	0.5811705
P2A	0.060058	13.1	0.7867598	13.7	0.8227946
FTA	0.035992	5.7	0.2051544	5.1	0.1835592
Total rebounds	-0.071093	7.8	-0.5545254	5.2	-0.3696836
Assists	-0.032491	10.2	-0.3314082	6.4	-0.2079424
Steals	0.536037	1.2	0.6432444	1.4	0.7504518
Blocks	0.325016	0.5	0.162508	0.5	0.162508
Personal fouls	0.262671	1.8	0.4728078	2.7	0.7092117
Linear score (log dds)	-3.037265		-2.6267159		-0.2613745
Odds	0.0480		0.0723		0.7700
Probability	**0.0458**		**0.0674**		**0.4350**

Conclusion & future directions

The findings not only support the correlation with physiological burden and knee injury risk in NBA players, as has been shown in other studies (Tummala et al., 2022), but additionally indicate that data on players' age and game-performance statistics can be used to build a prediction model for severe knee injury in NBA players. The model performance, though not very impressive, appears sufficiently accurate to be useful.

A future direction for this project is to try to improve the model's performance by adding predictors known to be associated with the risk of injury, such as the player's height and weight (or body mass index), the years of experience, and games missed due to injury (Tummala et al., 2022). Relevant data could be extracted from the same or other publicly available data sources.

Moreover, the model could potentially be refined by improving the modelling of features, such as the average minutes per game, that may have non-linear associations with the log odds for

severe knee injury. Another challenge that should be addressed is that the data points may not be completely independent, as there are multiple records per player in a series of seasons (here, we had n = 5010 records from 1290 different players).

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